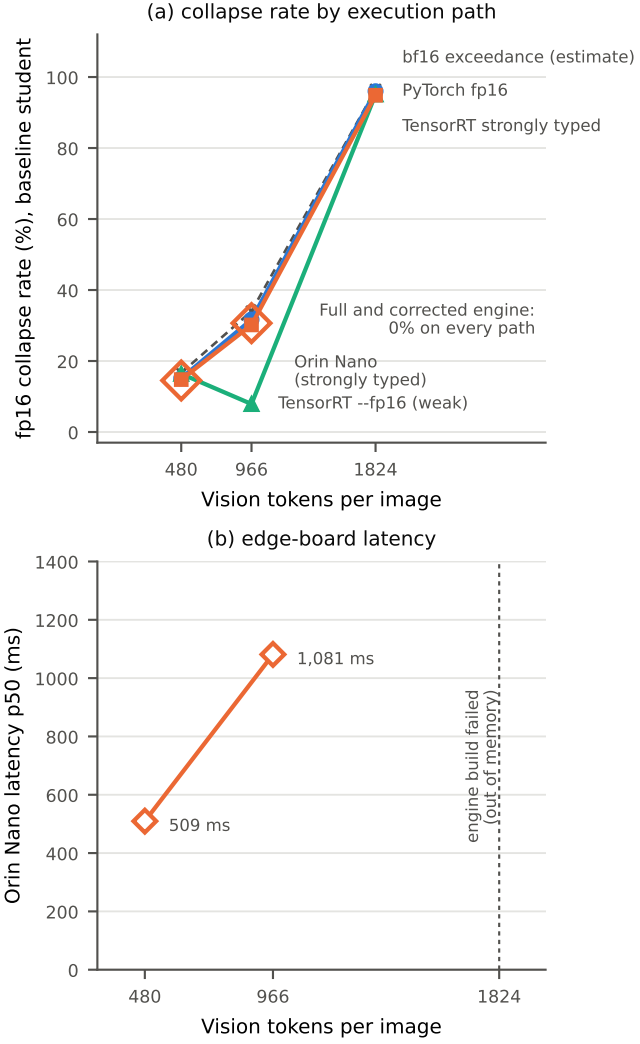
(a) collapse rate by execution path
fp16 collapse rate (%), baseline student
0
20
40
60
80
100
480
966
1824
Vision tokens per image
bf16 exceedance (estimate)
PyTorch fp16
TensorRT strongly typed
Full and corrected engine:
0% on every path
Orin Nano
(strongly typed)
TensorRT --fp16 (weak)
(b) edge-board latency
Orin Nano latency p50 (ms)
0
200
400
600
800
1000
1200
1400
480
966
1824
Vision tokens per image
509 ms
1,081 ms
engine build failed
(out of memory)